

DIYA BHATTARAI

Dallas-Fort Worth | diyabhattarai380@gmail.com | [linkedin.com/in/diya-bhattarai](https://www.linkedin.com/in/diya-bhattarai) | github.com/diyabh

Education

B.S., Computer Science

University of Texas at Dallas

2028

Richardson, TX

Skills

Languages: Python, Go, Java, JavaScript, TypeScript, C++, C#, Swift, SQL, HTML/CSS

Backend, AI & Data: FastAPI, React, Node.js, REST APIs, LLMs, Function Calling, Tool Calling, Machine Learning, PyTorch, TensorFlow, Databricks, Pandas, NumPy, Astropy, OpenCV, LLM APIs, Prompt Engineering, Retrieval Patterns

Systems & Tools: AWS, PostgreSQL, Supabase, Git/GitHub, GitHub Actions, CI/CD, SonarQube, Postman, Figma, AI Security

Computer Science & AI Fundamentals: Data Structures And Algorithms, Debugging, Software Testing, Feature Engineering, Model Evaluation

Experience

Humana

May 2026 - Aug 2026

AI / Full-Stack Engineer Intern

Remote / Louisville, KY

- Owned a GPT-4.1 claims analytics chatbot for Humana's Claims Control Tower, integrating LLM APIs to map natural-language questions to 32 controlled function calls over Databricks-backed claims, KPI, and drilldown data.
- Built the FastAPI orchestration layer with page-aware tool routing, multi-step execution, cached KPI context, and Server-Sent Events for grounded, real-time responses.
- Partnered with claims SMEs, PMs, engineers, and data architects to turn analyst workflows into chatbot-accessible KPI drilldowns for a platform tied to roughly \$800M in annual returned funds.
- Shipped production code with Databricks SQL integration, curated data tables, API handlers, SonarQube fixes, lint/CI pipeline work, debugging, and software testing reaching 80%+ coverage on new API/UI code.
- Demoed chatbot progress to SMEs and VP-level leaders; selected as the sole presenter from a 5-person hackathon team after earning runner-up, presenting to Humana's CIO, VPs, managers, and intern cohort.
- Chosen with the only intern project team to demo Loops and Claude developer workflows to 100+ employees, showing AI-assisted feature delivery from product ideation through coding, testing, and review.

Exoplanet Group

Apr 2025 - Present

Undergraduate Research Assistant

Richardson, TX

- Build AutoWISP photometry and calibration tools in Python using NumPy, Astropy, and OpenCV for light-curve extraction, calibration diagnostics, cloud screening, and researcher-facing BUI workflows.
- Manually label 502 images and apply feature engineering and model evaluation to star-loss, background-brightness, and red/blue color-ratio signals, improving cloudy-frame detection from calibration diagnostics.
- Contribute to AutoWISP v1.0.22, DOI: 10.5281/zenodo.20923935, documenting reproducible astronomical photometry and validation workflows.

Nebula Labs API

Aug 2025 - Present

Software Developer

Richardson, TX

- Develop a Go-based CLI for automated data collection and REST API ingestion, extending commands with goroutines, channels, validation, data structures and algorithms, tests, logging, error handling, and GitHub Actions CI.

ACM Research + HeXD Lab

Apr 2025 - Mar 2025

Undergraduate Researcher

Richardson, TX

- Build a YOLO-based XR perception system for real-time object detection, human-object tracking, and long-horizon video understanding.
- Generate temporal scene graphs with memory-aware retrieval patterns and connect Unity XR simulations to Python perception models for 3D spatial reasoning.

Replate / AIM Program

Jan 2025 - Apr 2025

Project Manager

Richardson, TX

- Lead a 6-member team building an AI-powered cooking assistant with live voice-and-camera guidance across CV, Flutter/mobile, backend, and SQL work.
- Plan milestones, coordinate integration, mentor teammates on GitHub and debugging, and run hands-on ML/CV workshops for 70+ students.

Code Ninjas

Feb 2023 - Jun 2025

Coding Tutor

Frisco, TX

- Taught Python, JavaScript, C#, debugging, and game-development fundamentals; led camps where students built Roblox games and Minecraft mods.

Projects & Awards

On the Grid

Nov 2025

Avangrid Hackathon

- Built a renewable-energy valuation framework by preprocessing 79,000+ hourly market records, applying feature engineering and Gradient Boosting, and evaluating pricing predictions with an R^2 of 0.95; awarded 2nd Place.

BreakPoint

Apr 2025

HackAI

- Developed an LLM-powered mental-health app using LLM APIs for real-time emotion and tone analysis, with secure authentication, encrypted journaling, offline storage, and backend persistence; awarded Most Creative Project at HackAI 2025.

OpticNet

Oct 2024

UTD AIMD

- Built and deployed a CNN for diabetic-retinopathy detection with image preprocessing, inference workflows, and model-interpretability visualizations; reached 93% validation accuracy and won 1st Place.